

A Model That Understands Your Medical Notes

Benjamin Shvartsman (bs828), Timothy Lin (tml4), Gaspard Loeillot (gl458)

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Application Setting: Patient- and staff-facing copilot over longitudinal patient notes, powered by MIMIC-III fine-tuned EmbeddingGemma model. We highlight four downstream potential applications: note search, evidence chat, free-text cohort discovery, and trajectory interpretation. Our research provides the substrate for intake understanding, evidence synthesis, and complex care workflows.

Project Description

General-purpose embedding models from providers like OpenAI encode text along many dimensions (style, syntax, topic), but they don't prioritize the clinical semantics that are critical in healthcare. Embeddings that faithfully capture clinical state rather than writing conventions might produce superior performance on downstream tasks like disease prediction, risk stratification, and survival analysis. Radical Health AI recently demonstrated this principle by fine-tuning EmbeddingGemma-300m on MIMIC-III patient notes using a temporal contrastive objective, achieving a 0.934 AUROC on diagnosis prediction versus 0.809 for OpenAI. We aimed to reproduce, validate, and extend that result.

Specifically, we attempted to replicate Radical Health's contrastive fine-tuning pipeline as a baseline, then systematically improve it along two axes:

1. Richer training signal through hierarchical contrastive objectives that incorporate structured clinical labels alongside the temporal note-pair signal.
2. Deployment as a production-ready embedding service on Baseten, integrated with Fasten Health's patient record aggregation layer to demonstrate real-world utility.

Our proposed final product resulting from these improvements was an end-to-end system from raw clinical text to actionable clinical intelligence.

Our project mostly adhered to this proposal. We were able to replicate the Radical Health contrastive fine-tuning pipeline on a subset of the MIMIC-III patient notes dataset and create a set of baseline embeddings (EmbeddingGemma-300m, Text Embedding 3 Small, Text Embedding 3 Large) to compare our models' performances to. We then systematically evaluated our models, generated artifacts, and deployed them to Hugging Face (rather than the originally proposed Baseten integration, we found this to be less suitable for our purposes and timeframe).

To demonstrate the practical utility of the fine-tuned embeddings, we built a staff-facing application layer over the indexed MIMIC corpus: a FastAPI backend and a single-page web frontend that together expose four downstream task modules.

Semantic note search ranks the ~23,000 anchor notes by cosine similarity over a FAISS index built from the hierarchical model's embeddings. Evidence-grounded Q&A extends this with retrieval-augmented generation: the top-k retrieved notes are passed to gpt-4o-mini under one of two prompting modes. In evidence mode, the model is allowed to synthesize patterns across cases and is required to produce [Note N] citations; in strict mode, it refuses to answer unless the literal answer appears in the retrieved text.

Cohort discovery supports two query types. The first is nearest-neighbor search over mean-pooled patient-level embeddings, seeded by a reference patient; alongside the ranked list, we report Jaccard overlap of ICD chapters as a quantitative sanity check. The second accepts a free-text phenotype description, for instance "elderly diabetic with acute kidney injury and respiratory failure," which we encode with the same fine-tuned model and rank against the patient-level vectors.

Patient trajectory analysis plots per-note embedding velocity (the L2 distance between consecutive note embeddings) together with a 2-D PCA projection of a patient's note sequence. Inflection points above a p95 spike threshold are flagged automatically, and an LLM generates a short natural-language interpretation that names those inflection points, summarizes the clinical course, and assigns one of five overall labels: stable, improving, declining, volatile, or mixed.

Collectively, these four modules are intended as the retrieval and reasoning substrate for downstream copilot workflows: longitudinal note review, intake note understanding, prior-authorization evidence assembly, and care-navigation onboarding for complex cases. The

common requirement across these settings is similarity search that is faithful to clinical content rather than to surface-level writing style.

The key aspects of AI at the core of our work are the fine tuning of an embedding model and its adaptation for modern AI usage frameworks such as RAG. In fine tuning, we focused primarily on implementing different loss functions. Following the Radical Health baseline, we fine tuned Google’s EmbeddingGemma-300m model using contrastive (InfoNCE) loss to identify, given an anchor note for some patient, the positive (i.e. a note for the same patient at a subsequent timestep) from a set of negatives drawn from other patients. InfoNCE forces the model to learn representations that capture the latent medical trajectory of a patient while becoming invariant to stylistic differences between clinicians, departments, or documentation templates.

To create richer training signal and account for the hierarchical structure of clinical labels, we then fine tuned EmbeddingGemma-300m using hierarchical contrastive loss instead to generate a set of embeddings that differ from each other depending on distance in the hierarchy rather than bluntly penalizing nonmatches (e.g., embeddings for ICD codes in the same category might differ less than those for ICD codes in different categories but the same chapter). We settled on using a HiMulCon (described by Zhang et al) style loss function for this purpose. We implemented more AI techniques in our application layer, which contains RAG and cohort discovery features.

Evaluation

This section evaluates model quality in three layers: (1) primary quantitative generalization performance, (2) embedding-geometry analysis (clinical content vs style), and (3) downstream application behavior (search/RAG, cohort discovery, trajectory). We present held-out metrics as the primary evidence and use in-sample diagnostic metrics as supporting context.

1. Evaluation Goals

In evaluating our model, we set out to answer three broad questions:

1. *How does the quality of embeddings for our hierarchical fine-tuned model compare with those for the InfoNCE and general purpose baselines?*
2. *Do our embeddings capture clinical content more than style?*
3. *How accurate are downstream tasks (search, cohort discovery, trajectory) when using our embeddings?*

2. Primary quantitative results: recall and diagnosis

In investigating the first question, we adopted an approach similar to Radical Health’s as one of our goals was to replicate their work. We first tested our general purpose models’ abilities to recall patient notes from the dataset using the top- k accuracy metric for $k=5, 10$. Next, we used the same testing methodology for our fine tuned InfoNCE and hierarchical models.

Additionally, we assessed the accuracy of our model for predicting diagnoses by using embeddings as input features for downstream clinical prediction tasks. We framed diagnosis prediction as a multi-label classification problem using ICD-9 diagnosis codes associated with each admission in MIMIC-III as ground truth labels and trained a OneVsRest logistic regression classifier on the 25 most frequent ICD codes. To assess accuracy, we calculated the macro AUROC (average of AUROC scores over all classes) for the classifiers trained on each model for a held out portion of the data.

Loss	Top-5 Recall (diagnostic) CI	Top-10 Recall (diagnostic) CI	Top-5 Recall (held-out) CI	Top-10 Recall (held-out) CI	Macro AUROC CI
Temporal InfoNCE	[46.51%, 47.74%]	[66.05%, 67.30%]	[16.11%, 17.19%]	[22.92%, 23.98%]	[0.8861, 0.9281]

Hierarchical (HiMulCon style)	[66.54%, 67.69%]	[84.00%, 84.89%]	[16.23%, 17.16%]	[23.03%, 24.09%]	[0.8827, 0.9256]
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Figure 1: The bootstrapped 95% confidence intervals for top-k and macro AUROC metrics evaluated from the fine-tuned InfoNCE and Hierarchical models.

Pretrained model	Top-5 (diagnostic)	Recall	Top-10 (diagnostic)	Recall	Macro AUROC
EmbeddingGemma-300m	5.99%		7.57%		88.52%
Text Embedding 3 Small	5.14%		6.44%		88.11%
Text Embedding 3 Large	5.45%		6.99%		89.31%

Figure 2: Models were additionally evaluated on a held-out (test) data set of the same size. Top-5, top-10, and macro AUROC metrics as evaluated on the held-out data are also given for the three baseline models.

Table placeholder (fine-tuned models): Top-5 Recall (diagnostic) CI, Top-10 Recall (diagnostic) CI, Top-5 Recall (held-out) CI, Top-10 Recall (held-out) CI, Macro AUROC CI

When evaluated on held out data, the top- k scores for both models decrease dramatically to a roughly similar level that is nevertheless higher than pretrained baselines.

3. Embedding geometry: style vs clinical content

For the second question, we adopted a mixture of quantitative and qualitative approaches. To quantify the degree to which our hierarchical and InfoNCE fine-tuned models captured clinical content over style or vice versa, we calculated silhouette scores, which measure the similarity of objects to their clusters (i.e., whether the classes of notes are more or less uniform within clusters). For each model, we calculated a silhouette score for style (which we deemed note

category to be a proxy for) and a silhouette score for clinical content (ICD-9 chapter). We subtract the latter from the former to produce a delta score where positive values indicate a geometry that separates more strongly by note style and negative values indicate one that separates more strongly by clinical content. We also used UMAP to visually inspect the clustering of our fine tuned models.

Figure 3:

Model	Silhouette (chapter)	Silhouette (category)	delta = cat – chap
OpenAI text-embedding-3-large	-0.045	+0.043	+0.089
OpenAI text-embedding-3-small	-0.054	+0.016	+0.070
google/embeddinggemma-300m (vanilla)	-0.053	-0.017	+0.036
InfoNCE fine-tuned	-0.057	-0.089	-0.032
Hierarchical fine-tuned	-0.066	-0.098	-0.032

Figure 3: Silhouette scores and deltas for all models.

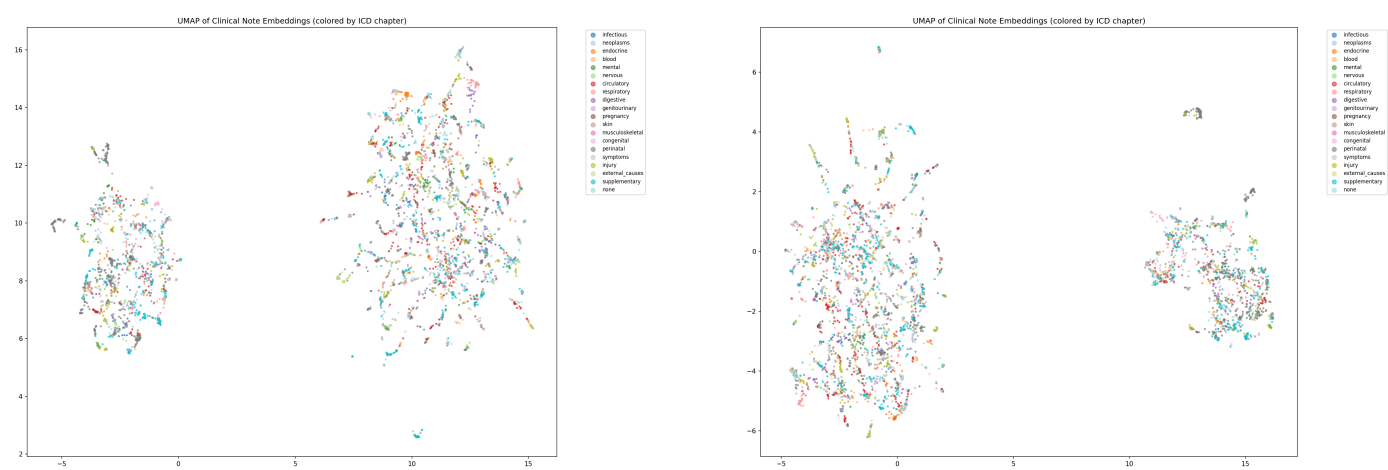


Figure 4: UMAP of note embeddings for hierarchical and InfoNCE models

Fine tuning modestly shifts the embeddings from organization by style (category) to clinical content (chapter). However, absolute silhouette scores for clinical content still remain negative. The embedding spaces for both models appear roughly similar. No widespread clean clustering of notes by ICD chapter is apparent. Clinical information could be encoded at a finer scale.

4. Downstream application evaluation

To evaluate practical utility beyond benchmark metrics, we tested the fine-tuned embeddings in four downstream workflows built on the same FAISS index over anchor-note embeddings: patient-record search, retrieval-augmented generation (RAG), cohort discovery, and longitudinal trajectory analysis. These downstream evaluations are intended as capability demonstrations of how the embedding space can be operationalized for clinical-text retrieval and analysis.

4.1 Patient-record search and retrieval-augmented generation (RAG)

We first evaluated whether free-text clinical queries retrieve semantically aligned notes from the indexed corpus. In this setup, queries are embedded with the same fine-tuned model used to embed the corpus, and nearest neighbors are returned by cosine similarity. Qualitatively, retrieved notes often reflect clinically coherent terminology and diagnosis context relevant to the query intent.

For RAG, top-k retrieved notes were passed to an LLM with instructions to answer only from retrieved evidence and refuse unsupported answers. This setup is useful for assessing whether the retrieval layer can provide enough relevant context for grounded synthesis, and whether the system can fail safely when evidence is insufficient.

```
=====
QUERY: newborn with respiratory distress requiring CPAP
=====
rank=1 sim=0.597 pid=8175 cat=Nursing/other hadm=155156 icd5=['7706', 'V290', '76519', '76527', 'V3000']
Neonatology Attending 33-6/7 week GA infant admitted with respiratory distress Maternal Hx - 39 year old G4P1->2 womanw ith POBHx notable f
rank=2 sim=0.587 pid=1743 cat=Radiology hadm=102748 icd5=['7706', 'V290', 'V053', '7488', '76519']
[**2152-6-20**] 5:56 AM BABYGRAM (CHEST ONLY) Clip # [**Clip Number (Radiology) 15634**] Reason:
rank=3 sim=0.573 pid=8462 cat=Nursing/other hadm=152811 icd5=['V053', '77081', '76525', 'V290', '7793']
Neonatology Attending - Addendum Called to evaluate infant for increased FiO2 requirement. Infant on CPAP6, 43%. Int tachypneic to 70s. No
rank=4 sim=0.548 pid=6642 cat=Nursing/other hadm=115750 icd5=['V290', 'V3001', '76517', '7742', '76527']
Neonatology Attending Admission Note Infant placed on CPAP for respiratory distress. CXR c/w TTN rather than RDS or pneumonia. No air leak.
rank=5 sim=0.548 pid=3144 cat=Radiology hadm=179849 icd5=['V3101', '76519', '769', 'V290', '76528']
[**2127-7-1**] 8:39 PM BABYGRAM (CHEST ONLY) Clip # [**Clip Number (Radiology) 50485**] Reason:
```

```

=====
QUESTION: For the newborn with respiratory distress requiring CPAP, what respiratory findings and immediate management steps are documented in the retrieved notes?
=====

[gpt-4o-mini answer]
The respiratory findings documented for the newborn with respiratory distress requiring CPAP include:

1. **Mild retractions** and **grunting respirations** prior to CPAP (Note 2).
2. **Mild work of breathing** and **subcostal retractions** while on CPAP (Note 4).
3. **Diminished air exchange bilaterally** noted during the physical examination (Note 4).
4. **Diffuse hazy opacity of the lungs** consistent with mild hyaline membrane disease or pulmonary edema observed on chest imaging (Note 3).

Immediate management steps documented include:

1. **Oral and nasal bulb suctioning** and **drying** of the infant (Note 2).
2. **Free flow oxygen administration** for 2 minutes (Note 2).
3. **Brief facial CPAP** applied for subsequent mild retractions and grunting (Note 2).
4. The infant was placed on CPAP for respiratory distress (Note 5).

These findings and management steps illustrate the immediate response to the newborn's respiratory distress.

Top-3 retrieved notes:
Note 1 sim=0.669 pid=1743 cat=Radiology icd5=['7706', 'V290', 'V053', '7488', '76519']
Note 2 sim=0.616 pid=8175 cat=Nursing/other icd5=['7706', 'V290', '76519', '76527', 'V3000']
Note 3 sim=0.584 pid=3144 cat=Radiology icd5=['V3101', '76519', '769', 'V290', '76528']

```

Figure 6: Query-aligned patient-record retrieval and grounded RAG response (newborn respiratory distress case)

4.2 Cohort discovery via patient-level similarity

We next evaluated patient-level similarity by mean-pooling note embeddings per patient and ranking nearest neighbors in embedding space. To quantify whether nearest-neighbor patients are clinically more related than random pairs, we used ICD-chapter Jaccard overlap and a one-sided Mann-Whitney test.

This provides a task-oriented sanity check that geometric proximity corresponds to meaningful diagnostic overlap at the chapter level. In our experiments, nearest-neighbor pairs showed higher chapter overlap than random baselines, supporting that the embedding space captures patient-level clinical affinity.

```

Patient-level embeddings: (483, 768)

=== seed subject_id=7627 (n_anchors=747) ===

```

	rank	similarity	subject_id	shared_chapters	n_shared	jaccard_chapter
0	1	0.304613	6766	[740-759_congenital, 760-779_perinatal, V_supplementary]	3	0.333333
1	2	0.279640	8464	[460-519_respiratory, 520-579_digestive, 780-799_symptoms]	3	0.214286
2	3	0.265282	9212	[740-759_congenital, 760-779_perinatal, V_supplementary]	3	0.375000
3	4	0.229894	7403	[V_supplementary]	1	0.142857
4	5	0.227494	9294	[460-519_respiratory]	1	0.100000

```
{
  "n_seeds": 50,
  "k": 5,
  "n_random_pairs": 1000,
  "neighbor_mean_jaccard": 0.45130021645021645,
  "neighbor_median_jaccard": 0.4226190476190476,
  "random_mean_jaccard": 0.21549398379398377,
  "random_median_jaccard": 0.18181818181818182,
  "lift_mean": 2.09425900670001,
  "mannwhitneyu_one_sided_p": 4.769717015804531e-41
}
```

Figure 7: Cohort discovery with patient-level embeddings: nearest-neighbor examples and ICD-chapter overlap validation

4.3 Trajectory analysis and embedding velocity

Finally, we examined within-patient temporal dynamics using note-to-note embedding velocity (L2 distance to previous note embedding) and 2D PCA trajectories. This analysis is designed to test whether the embedding sequence is temporally non-stationary and whether high-velocity transitions can identify candidate inflection points for review.

The trajectory visualization shows nontrivial movement through embedding space over note order, and velocity plots reveal localized spikes relative to each patient’s baseline. We also compared velocity distributions across patients to show that temporal embedding dynamics are patient-specific rather than uniform across the corpus.

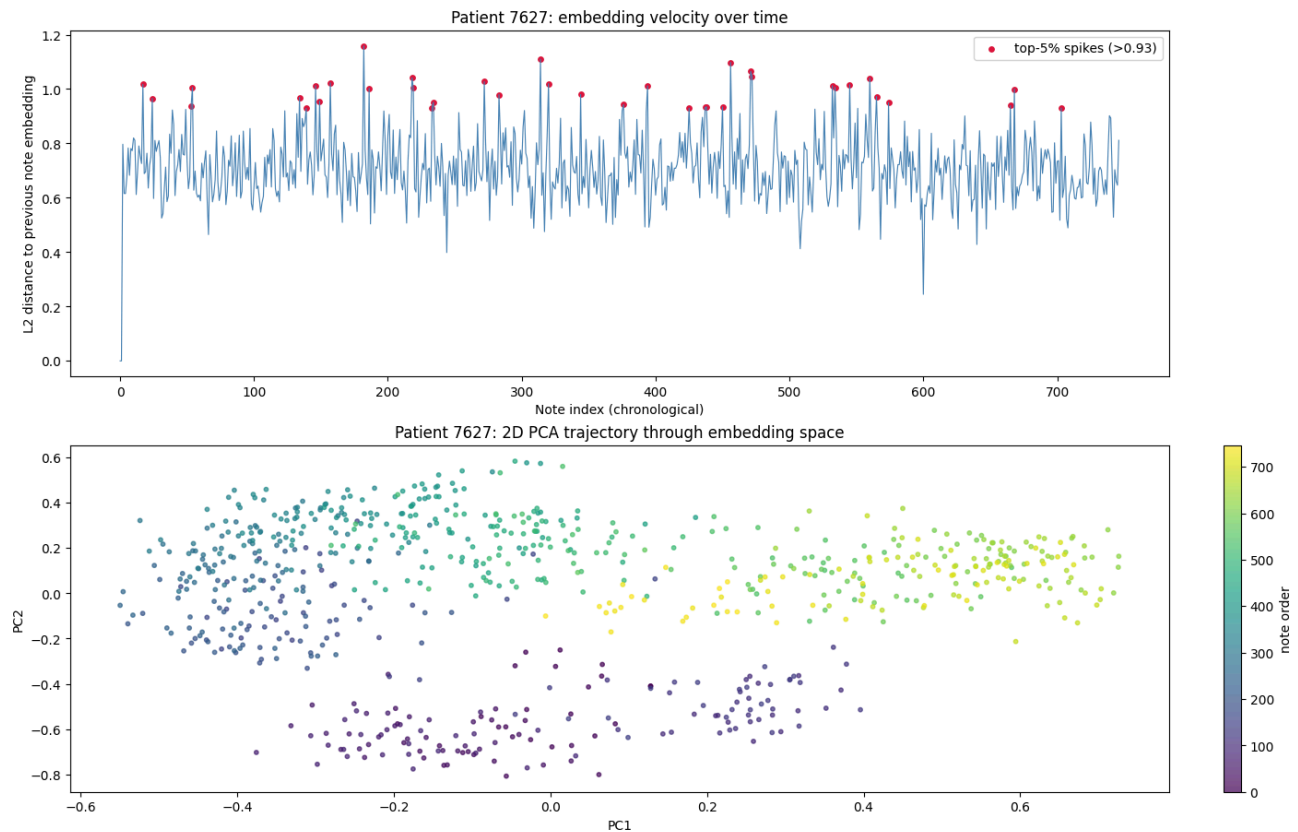


Figure 8: Patient trajectory in embedding space: note-to-note embedding velocity over time (top) and 2D PCA trajectory colored by note order (bottom)

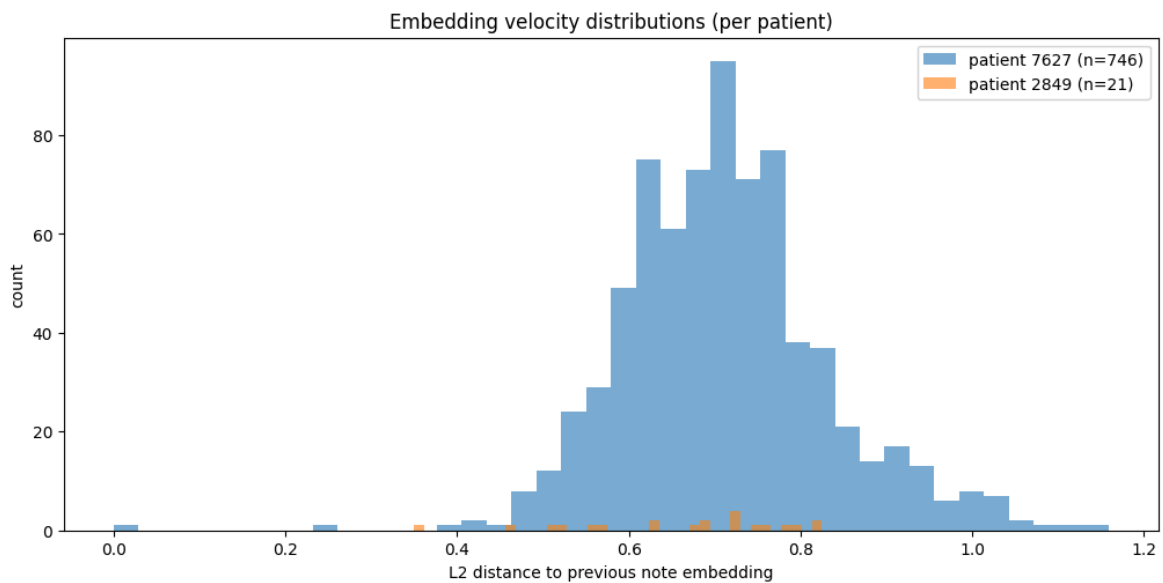


Figure 9: Embedding velocity distributions for two patients (L2 distance to previous note embedding)

To avoid over-claiming, we treat this trajectory analysis as exploratory: it demonstrates temporal sensitivity of the embedding representation and provides candidate transition points, but does not by itself establish causal linkage to ground-truth clinical events.

4.4 Summary of downstream findings

Across these downstream tasks, the fine-tuned embedding space supports practical retrieval and analysis behaviors: semantically coherent note retrieval, RAG workflows with grounded/refusal behavior, patient-level cohort similarity signal, and interpretable within-patient temporal dynamics. Together, these results suggest that the embeddings are not only improved on benchmark-style retrieval metrics, but also usable as a foundation layer for clinical-text application patterns.

5 Conclusion and limitations

Though we did not meet our proposed goal of a 65% top-5 accuracy comparable to Radical Health’s model, we still consider our work largely successful given our findings and practical limitations.

We have demonstrated that fine tuning the EmbeddingGemma-300m model using hierarchical contrastive learning objectives can produce slight improvements over other fine-tuned and pre-trained baselines. The resulting embeddings and their distribution are meaningful and reveal a shift to a geometry that better captures clinical content rather than style. Moreover, we have demonstrated several useful downstream applications using our model.

The most significant limitation of our work is the size of our true training dataset: 23,657 notes associated with 500 patients (the Kaggle dataset includes 10,000 patients and is itself only a subset of the full dataset, which we could not access). We determined that training and embedding the notes of thousands of patients would have been too impractical given the compute available to us and our prioritization of rapid development. A complete run of our pipeline using a T4 GPU and all available data would have taken a full day or more. It is possible, however, that using significantly more data for training could have produced more robust embeddings.

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